

Econ 139: Labor Economics

The Current Population Survey (CPS)

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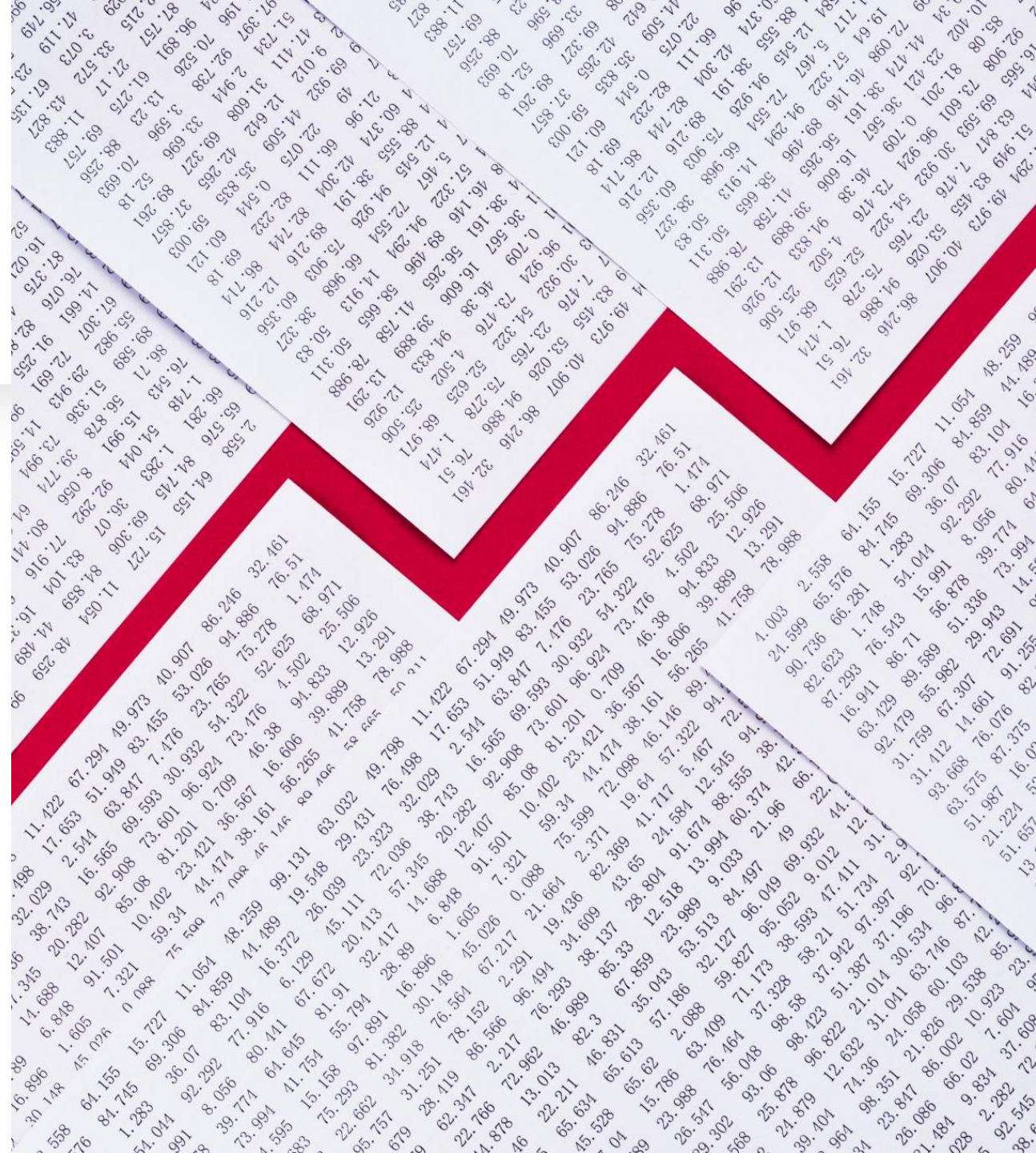


**U.S. BUREAU
OF LABOR
STATISTICS**

**United StatesTM
Census
Bureau**

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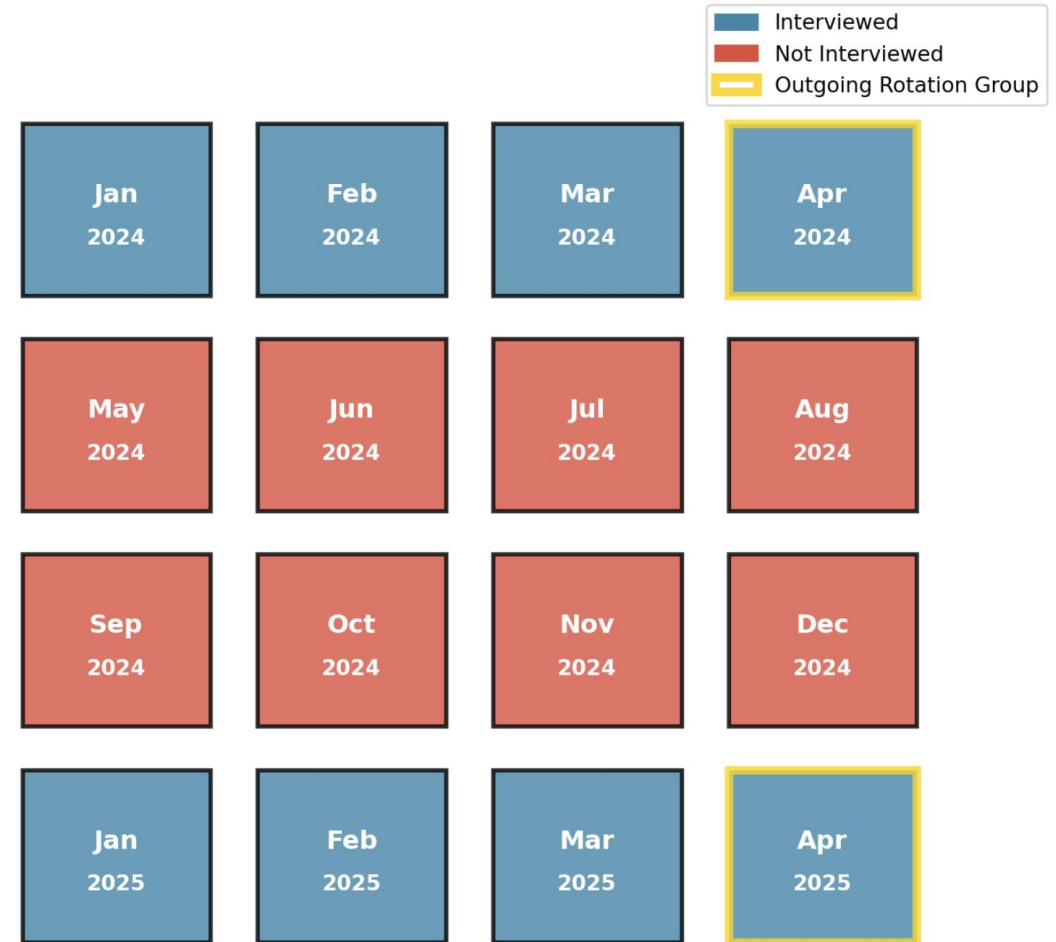
- The CPS is a large-scale survey undertaken by the U.S. Census Bureau and the Bureau of Labor Statistics
- Surveys 60,000 households per month
- Asks questions about labor force participation, demographics, work, among others
 - In certain months, there are also supplements that study particular topics



CPS Structure

- The CPS is a rotating panel
- The 4th and 8th month an individual is in the sample are referred to as outgoing rotation groups
 - These individuals are asked additional questions about their income
- The variable MISH indicates the interview round for the individual

Current Population Survey Design (Household First Interviewed in January 2024)



Downloading the CPS

- Two basic ways to download the CPS
 - Bureau of Labor Statistics makes files available at their website
 - IPUMS-CPS is a project that provides a user-friendly interface to select variables and samples. This means if you are using data over a long horizon, you don't have to manually combine and harmonize variables yourself
 - Let's take a look at IPUMS now

Computing the Unemployment Rate

- One of the uses of the CPS is to provide up-to-date information on the labor force
- Our first exploration of the dataset will be to compute the unemployment rate
 - The CPS is generally the source of the unemployment rate you read about in the news
- On the Canvas website I have added a file `cps_aug_25.csv` that contains a small subset of variables available for the August 2025 CPS

Software in this Class

- We will have a number of classes and assignments where we explore data
- I'll allow most statistical software to be used in the course, but you will need to submit either the excel spreadsheet you use or the code
- The most popular I expect will be Excel or R, but you can also use Stata or Python if you like
- Excel is still probably the most widely used in most industry settings. If you are interested in learning how to code, I would recommend using either R or Python. Today we will do the analysis in both Excel and R. If you are interested in graduate school in Economics, Stata would also be relevant

Variables

Table 7.1: CPS Dataset Variables (Total: 23 variables).

YEAR	PERNUM	SEX	WHYUNEMP
SERIAL	WTFINL	EMPSTAT	EMPSAME
MONTH	CPSIDP	LABFORCE	EDUC
HWTFINL	CPSIDV	OCC	UNION
CPSID	EARNWEEK2	CLASSWKR	UHRSWORKORG
MISH	AGE	UHRSWORKT	

Distribution of LABFORCE

Table 7.2: Distribution of Labor Force Status (LABFORCE Variable).

Labor Force Status	Description	Count	Percentage
0	NIU (Not in Universe)	16167	16.54%
1	No, not in labor force	32988	33.75%
2	Yes, in labor force	48585	49.71%

Variables in CPS are often categorical. Data is stored as a number (0,1,2), with each number signifying something different. An easy way to figure out the labels is google “varname CPS-IPUMS”. For example “LABFORCE CPS-IPUMS”



NIU

- NIU means Not-in-universe
- For some reason this individual was not asked this question
- For LABFORCE this is usually because the individual is younger than 16 years old (parents can fill out the current population survey for their children)

Distribution of LABFORCE (Dropping Missing Individuals)

Table 7.3: Distribution of Labor Force Status (Ages 16+).

Labor Force Status	Description	Count	Percentage
1	No, not in labor force	31929	39.72%
2	Yes, in labor force	48448	60.28%

EMPSTAT gives us a more detailed breakdown



Table 7.4: Distribution of Employment Status (EMPSTAT Variable).

Employment Status	Description	Count	Percentage
10	At work	44348	55.17%
12	Has job, not at work last week	2069	2.57%
21	Unemployed, experienced worker	1815	2.26%
22	Unemployed, new worker	216	0.27%
32	Not in labor force, unable to work	3704	4.61%
34	Not in labor force, other	9190	11.43%
36	Not in labor force, retired	19035	23.68%

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$$\begin{aligned}\text{Unemployment Rate} &= \frac{\text{Number of Unemployed}}{\text{Labor Force}} \\ &= \frac{2031}{2031 + 46417} \\ &= 4.4\%\end{aligned}$$

WTFINAL

- The Unemployment Rate we retrieved is very close to official statistics, but the official statistics use weights
- The reason is that the CPS does not randomly sample individuals
 - Some individuals are more likely to be chosen than others based on demographics
 - Downweight individuals that are “oversampled” and upweight individuals that are “undersampled”
- Important for being precise, but we are not going to use the weights in this class



Occupation

- As part of the Occupation Presentation, you will need to collect information from the CPS on your occupation
- Let's look at the 5 most common occupations

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Table 7.5: Top 5 Most Common Occupations Among Employed Workers

Occupation Code	Occupation Title	Count	Percentage
440	Managers, all other	1707	3.68%
3255	Registered Nurses	1035	2.23%
9130	Unknown Occupation	1003	2.16%
2310	Elementary and middle school teachers	959	2.07%
4700	First-line supervisors of retail sales workers	893	1.92%

EARNWEEK2

- EARNWEEK2 reports the average weekly earnings of individuals rounded
 - Until recently, earnings were not rounded
- The missing code for EARNWEEK2 is 999999.99. If you didn't realize this imagine how confusing this might be

Table 7.6: Distribution of EARNWEEK2 = 999999.99 (Not in Universe).

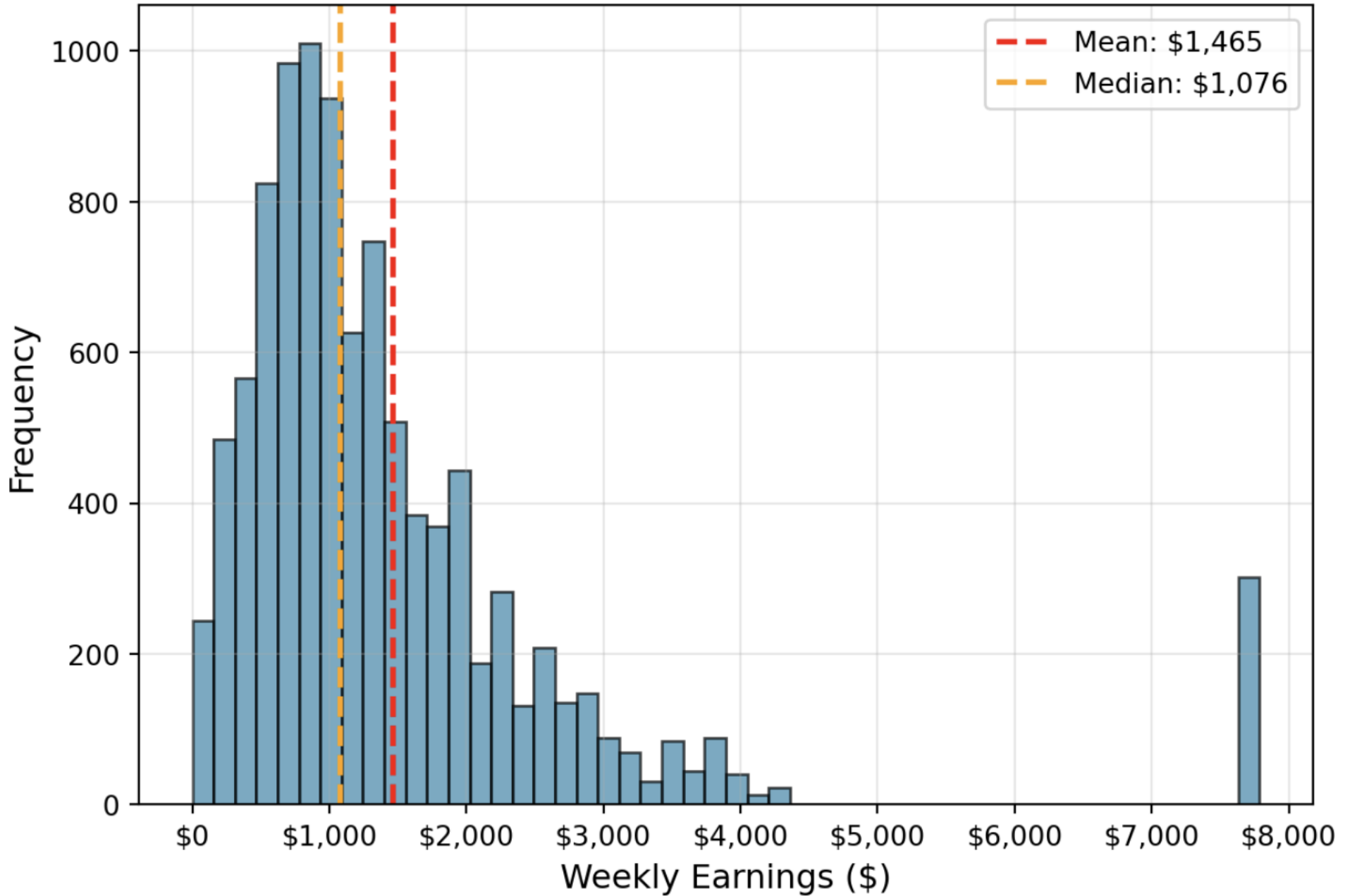
EARNWEEK2 Value	Count	Percentage
999999.99	70372	87.55%
All Other Values	10005	12.45%

EARNWEEK2

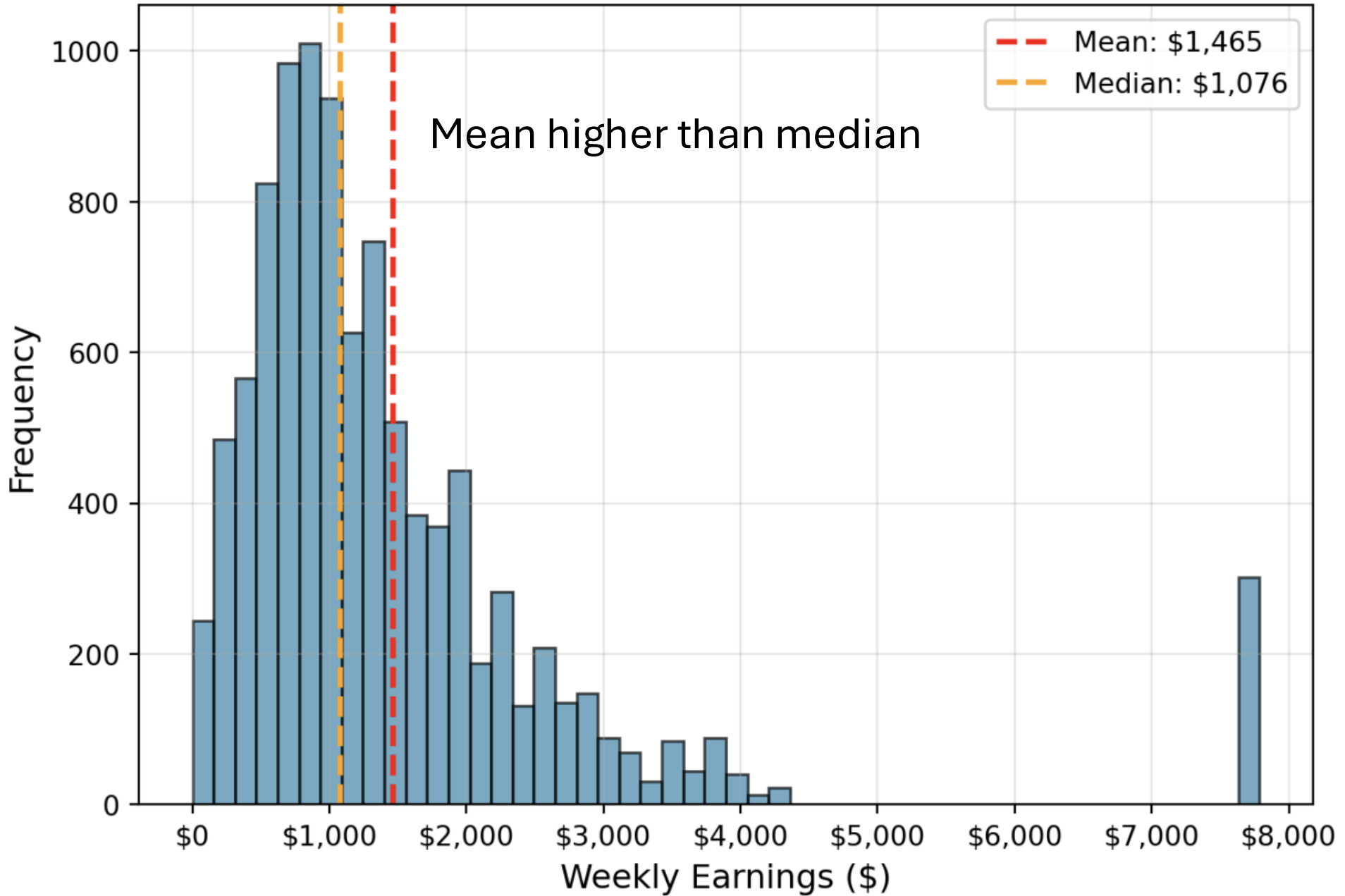
Table 7.7: Weekly Earnings Summary Statistics (Excluding Missing Values).

Statistic	Value
Mean Weekly Earnings	\$1,464.58
Median Weekly Earnings	\$1,076.00
10th Percentile	\$390.00
90th Percentile	\$2,700.00
Observations	10,005

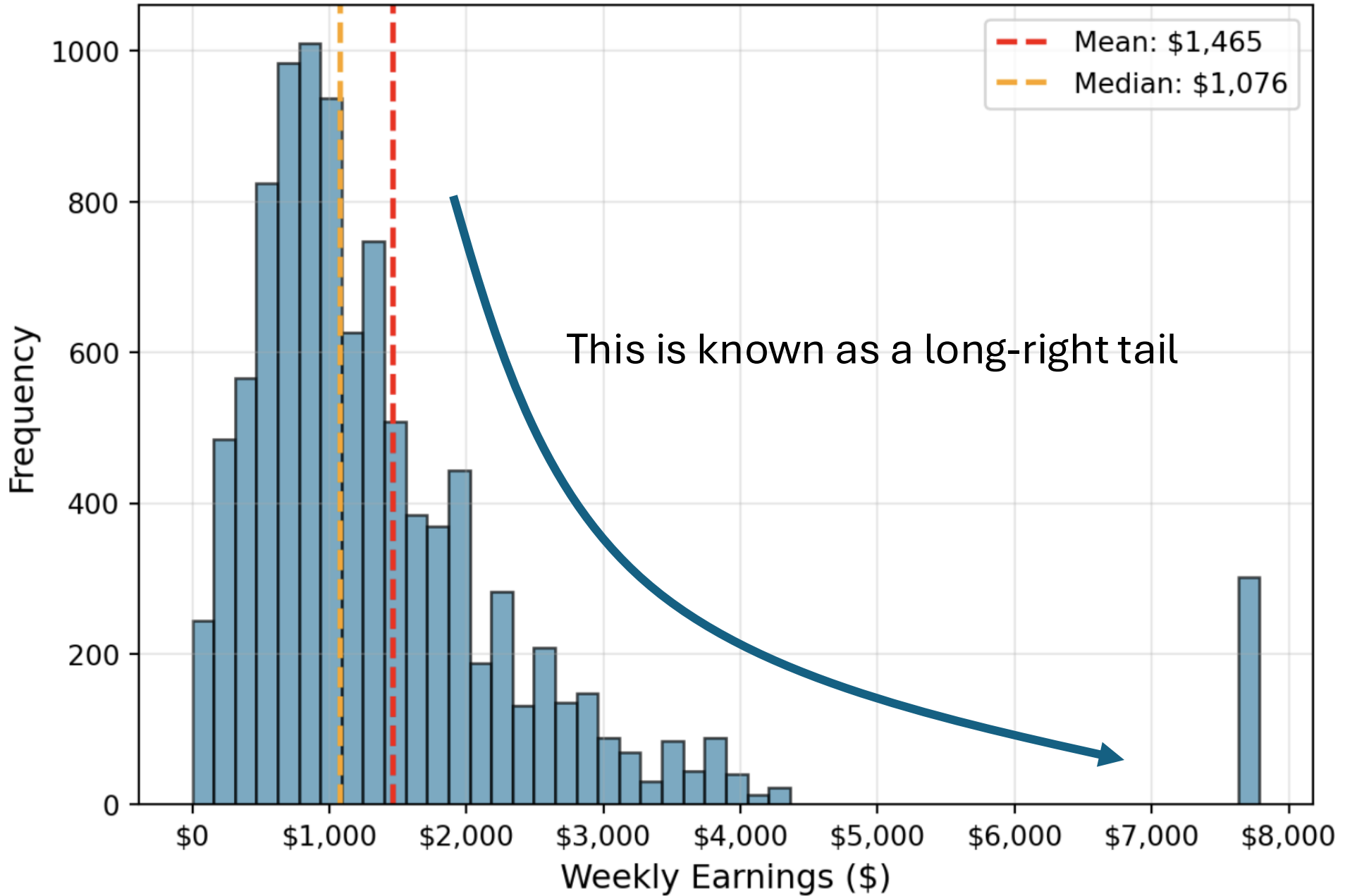
Distribution of Weekly Earnings (CPS August 2025)



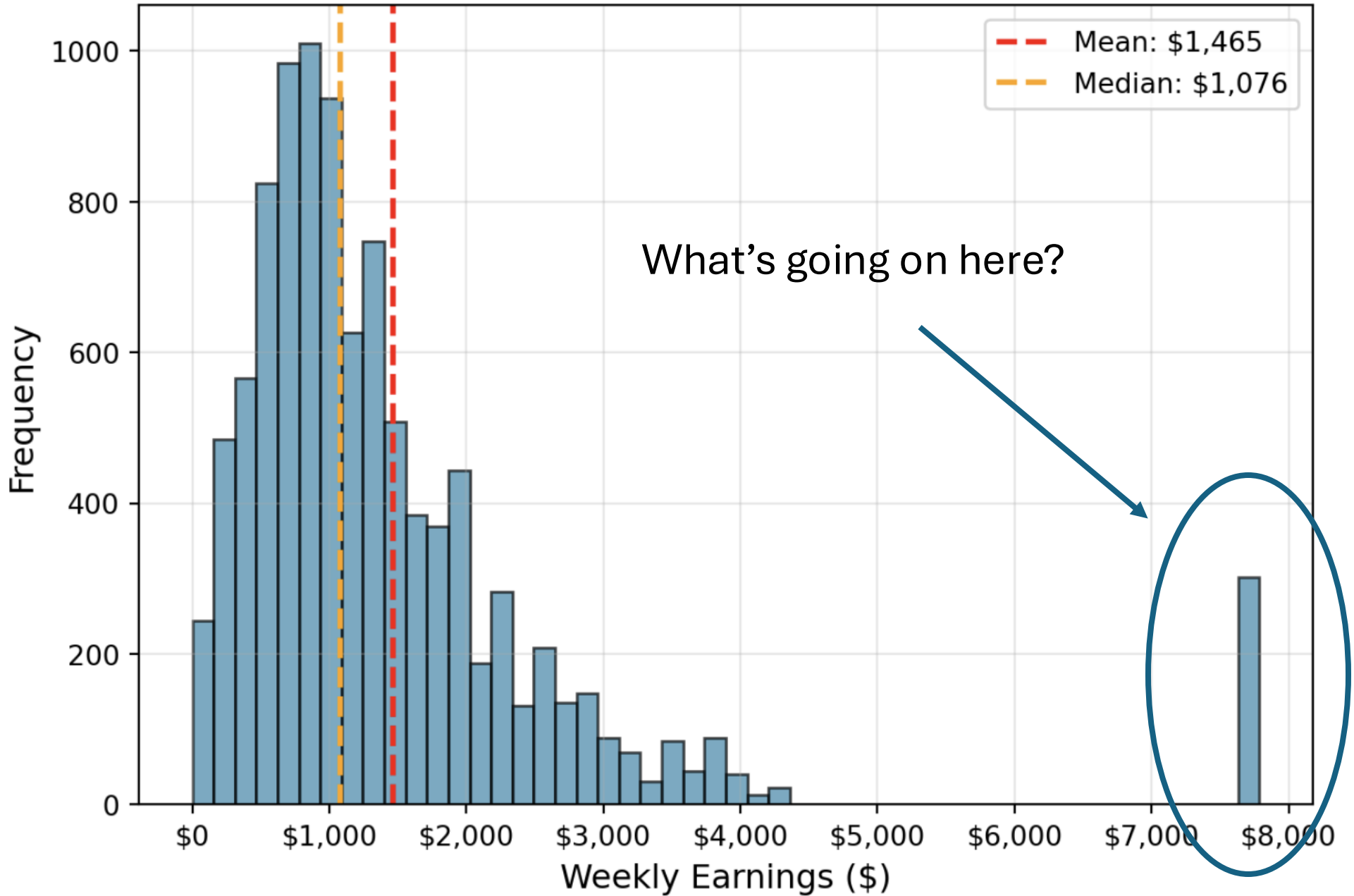
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Distribution of Weekly Earnings (CPS August 2025)



Top-coding

- CPS censors very high incomes to avoid confidentiality concerns
 - In the past, it was a fixed number
 - Now it is the top 3 percent of earners
- These censored observations are assigned the average income of everyone in the top 3 percent
 - That is why there are no observations above \$4,300 a week but a huge spike at \$7,785.88 (the average among the top 3 percent)

The CPS

- We have only looked at a small subset of the data, but already, we needed to be very careful to get the right analysis
 - Some individuals are not asked every question and this can be for multiple reasons
 - Some questions have potentially complicated categorical groups
 - Some questions have changed over time
- Large-scale data is extremely useful. CPS basically started modern labor economics in the US. But it requires careful data analysis to be useful